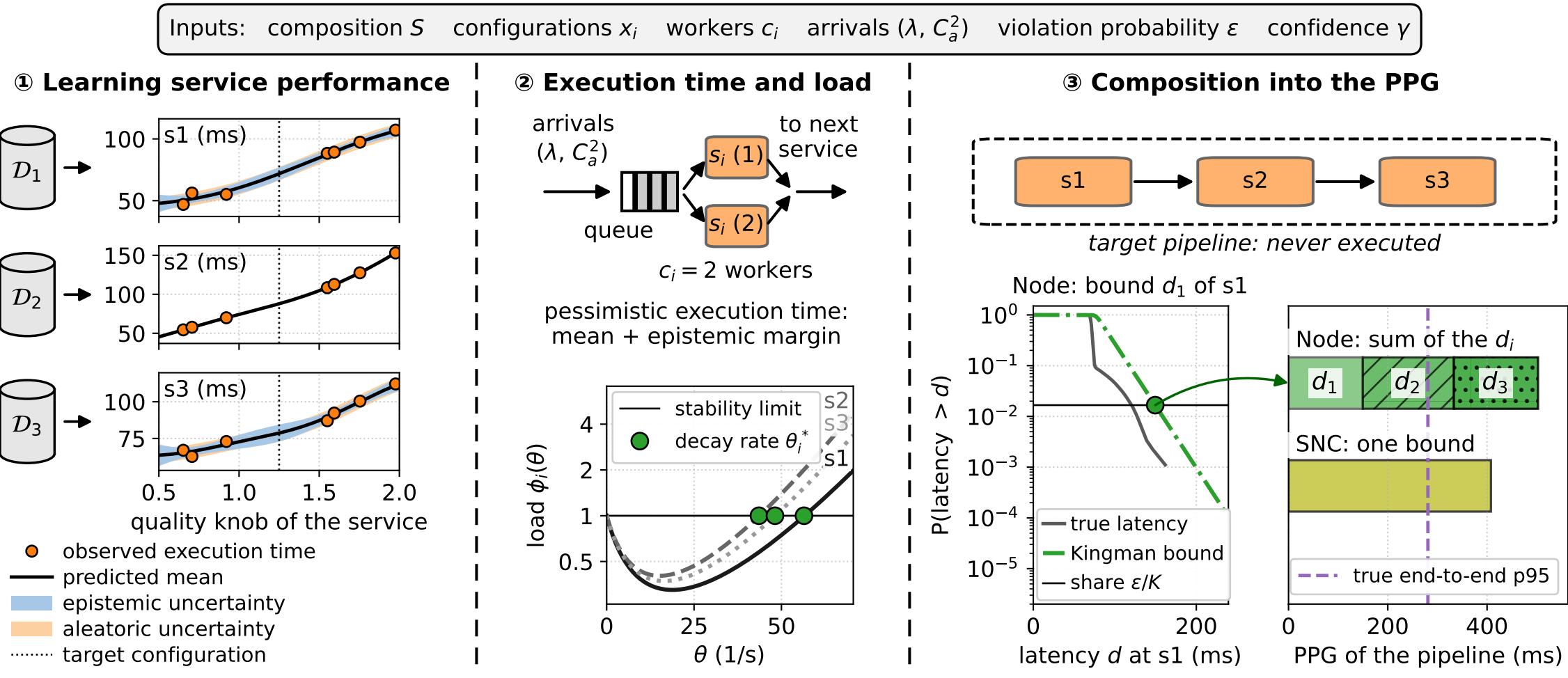

Inputs: composition S configurations x_i workers c_i arrivals (λ, C_a^2) violation probability ε confidence γ
① Learning service performance
$\mathcal{D}_1$
$\mathcal{D}_2$
$\mathcal{D}_3$
s1 (ms)
s2 (ms)
s3 (ms)
100
50
150
100
50
100
75
0.5
1.0
1.5
2.0
quality knob of the service
observed execution time
predicted mean
epistemic uncertainty
aleatoric uncertainty
target configuration
② Execution time and load
arrivals (λ, C_a^2)
queue
s_i (1)
s_i (2)
to next service
$c_i = 2$ workers
pessimistic execution time: mean + epistemic margin
stability limit
decay rate θ_i^*
s2
s3
s1
load $\phi_i(\theta)$
4
2
1
0.5
0
25
50
θ (1/s)
③ Composition into the PPG
s1
s2
s3
target pipeline: never executed
Node: bound d_1 of s1
10^0
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
P(latency > d)
true latency
Kingman bound
share ε/K
0
200
latency d at s1 (ms)
Node: sum of the d_i
d_1
d_2
d_3
SNC: one bound
true end-to-end p95
0
200
400
PPG of the pipeline (ms)